NATIONAL UNIVERSITY OF SINGAPORE

SCHOOL OF COMPUTING

MIDTERM ASSESSMENT FOR
Semester 2 AY2019/20

CS3244: MACHINE LEARNING

March 26, 2020 Time Allowed: 1 Hour 30 Minutes

INSTRUCTIONS TO CANDIDATES

1. This assessment paper contains **FOUR (4)** parts and comprises **ELEVEN (11)** printed pages, including this page.
2. Answer **ALL** questions as indicated.
3. This is a **RESTRICTED OPEN BOOK** assessment.
4. Please write your **Student Number** below. Do not write your name.

STUDENT NUMBER: _____

EXAMINER'S USE ONLY		
Part	Mark	Score
I	20	
II	15	
III	12	
IV	13	
TOTAL	60	

In Part I, II, III, and IV, you will find a series of structured questions. For each structured question, give your answer in the reserved space in the script.

Part I
Concept Learning 1

(20 points) Structured questions. Answer in the space provided on the script.

1. (10 points) Let $Z = \{0, 1, \ldots, 10\}$. Consider the input instance space $X = \{(x_1, x_2)\}_{x_1, x_2 \in Z}$ consisting of integer points in the x_1, x_2 plane, and the hypothesis space H such that each hypothesis $h \in H$ is defined as

$$h(x_1, x_2) = \begin{cases} 1 & \text{if } a \leq x_1 \leq b \text{ and } c \leq x_2 \leq d, \\ 0 & \text{otherwise;} \end{cases}$$

where $a, b, c, d \in Z$. We represent hypothesis h in the form (a, b, c, d). For example, a typical hypothesis in H is $(3, 5, 2, 9)$. Note that for any $h = (a, b, c, d) \in H$, if $a > b$ or $c > d$, then **no** input instance $(x_1, x_2) \in X$ satisfies h.

Trace the CANDIDATE-ELIMINATION algorithm (reproduced below in Fig. 1) for the hypothesis space H given the sequence of positive ($c(x_1, x_2) = 1$) and negative ($c(x_1, x_2) = 0$) training examples from Table 1 below (i.e., show the sequence of S and G boundary sets). You only need to show the **semantically distinct**[1] hypotheses in each boundary set.

1. $G \leftarrow$ maximally general hypotheses in H
2. $S \leftarrow$ maximally specific hypotheses in H
3. For each training example d
   - If d is a positive example
     - Remove from G any hypothesis inconsistent with d
     - For each $s \in S$ not consistent with d
       - Remove s from S
       - Add to S all minimal generalizations h of s s.t.
         h is consistent with d, and
         some member of G is more general than h
       - Remove from S any hypothesis that is more general than another hypothesis in S
   - If d is a negative example
     - Remove from S any hypothesis inconsistent with d
     - For each $g \in G$ not consistent with d
       - Remove g from G
       - Add to G all minimal specializations h of g s.t.
         h is consistent with d, and
         some member of S is more specific than h
       - Remove from G any hypothesis that is more specific than another hypothesis in G

Figure 1: CANDIDATE-ELIMINATION algorithm.

Example	Input Instance		Target Concept
	x_1	x_2	$c(x_1, x_2)$
1	6	3	1
2	8	7	0
3	4	7	1
4	2	1	0
5	3	9	0

Table 1: Positive ($c(x_1, x_2) = 1$) and negative ($c(x_1, x_2) = 0$) training examples for target concept c.

[1] The notion of **semantically distinct** hypotheses was mentioned and explained informally on page 10 of the "Concept Learning" lecture slides. For a formal definition, continue reading. Let hypotheses h and h' be in the same hypothesis space H, i.e., $h, h' \in H$. We know that a hypothesis represents a set of input instances in X such that every input instance in this set satisfies this hypothesis.

The **difference** $h \setminus h'$ of the hypotheses h and h' is defined as $h \setminus h'(x) = ((h(x) = 1) \wedge (h'(x) = 0))$ for all $x \in X$ and therefore represents the set difference of the sets of input instances represented by h and h'.

The hypotheses h and h' are **semantically distinct** iff there exists some $x \in X$ satisfying $h \setminus h'$ or $h' \setminus h$, that is, $\exists x \in X \;\; (h \setminus h'(x) = 1) \vee (h' \setminus h(x) = 1)$.

Solution:

$G_0 = \{(0, 10, 0, 10)\}$

$S_0 = \{(6, 5, 3, 2), (7, 4, 4, 1), (10, 0, 10, 0), \ldots\} = \{(6, 5, 3, 2)\}$

The above hypotheses in S_0 are **not semantically distinct.**

$G_1 =$

$S_1 =$

$S_2 =$

$G_2 =$

$G_3 =$

$S_3 =$

$S_4 =$

$G_4 =$

$S_5 =$

$G_5 =$

2. (10 points) Let $Z = \{0, 1, \ldots, 10\}$. Consider the input instance space $X = \{(x_1, x_2)\}_{x_1, x_2 \in Z}$ (i.e., same as that in the above question 1) consisting of integer points in the x_1, x_2 plane, and a **new hypothesis space** H' such that each hypothesis $h \in H'$ is defined as

$$h(x_1, x_2) = \begin{cases} \mathbf{0} & \textbf{if } a \leq x_1 \leq b \text{ and } c \leq x_2 \leq d, \\ \mathbf{1} & \textbf{otherwise}; \end{cases}$$

where $a, b, c, d \in Z$. We represent hypothesis h in the form (a, b, c, d). For example, a typical hypothesis in H' is $(3, 5, 2, 9)$. Note that for any $h = (a, b, c, d) \in H'$, if $a > b$ or $c > d$, then **every** input instance $(x_1, x_2) \in X$ satisfies h.

Trace the CANDIDATE-ELIMINATION algorithm (reproduced above in Fig. 1) for the hypothesis space H' given the sequence of positive ($c(x_1, x_2) = 1$) and negative ($c(x_1, x_2) = 0$) training examples from **Table 2** below (i.e., show the sequence of S and G boundary sets). You only need to show the **semantically distinct** hypotheses in each boundary set.

Example	Input Instance		Target Concept
	x_1	x_2	$c(x_1, x_2)$
1	3	6	0
2	7	8	1
3	7	4	0
4	1	2	1
5	9	3	1

Table 2: Positive ($c(x_1, x_2) = 1$) and negative ($c(x_1, x_2) = 0$) training examples for target concept c.

Solution:

$G_0 = \{(6,5,3,2),(7,4,4,1),(10,0,10,0),\ldots\} = \{(6,5,3,2)\}$

$S_0 = \{(0,10,0,10)\}$

The above hypotheses in G_0 are **not semantically distinct.**

$S_1 =$

$G_1 =$

$G_2 =$

$S_2 =$

$S_3 =$

$G_3 =$

$G_4 =$

$S_4 =$

$G_5 =$

$S_5 =$

Part II
Concept Learning 2

(15 points) Structured questions. Answer in the space provided on the script.

Let G and S be the general and specific boundaries of the version space $VS_{H,D}$, respectively.

1. (10 points) Prove formally or disprove that for any $h \in H$, if none of $g \in G$ is *more general than or equal to* h, then h is not *more general than or equal to* any $s \in S$, that is,

$$\forall h \in H \quad (\forall g \in G \;\; g \not\geq_g h) \longrightarrow (\forall s \in S \;\; h \not\geq_g s) \; .$$

Hint: You may wish to consider using the content of question 1 in Part I to help you establish an informal intuition for solving this question.

Solution:

Solution:

2. (5 points) Prove formally or disprove that for any $\boldsymbol{h} \in \boldsymbol{VS_{H,D}}$, if none of $g \in G$ is *more general than or equal to* h, then h is not *more general than or equal to* any $s \in S$, that is,

$$\forall \boldsymbol{h} \in \boldsymbol{VS_{H,D}} \quad (\forall g \in G \;\; g \not\geq_g h) \longrightarrow (\forall s \in S \;\; h \not\geq_g s) \; .$$

Hint: You may wish to consider using the Version Space Representation Theorem (page 20 of the "Concept Learning" lecture slides):

$$VS_{H,D} = \{h \in H \mid \exists s \in S \; \exists g \in G \quad g \geq_g h \geq_g s\} \; .$$

Solution:

Part III
Decision Tree Learning

(12 points) Structured questions. Answer in the space provided on the script.

In this question, consider the hypothetical task of learning the target concept $MLGrade$ with a *decision tree* to understand the factors affecting the grades of students enrolled in an ML class using the 14 training examples listed in Table 3 below.

Example Student	Input Instances				Target Concept
	$AttendClass$	$MidtermGrade$	$ProjectGrade$	$LoveML$	$MLGrade$
$Mohit$	$Always$	$Poor$	$Average$	Yes	$Fail$
$ChiThanh$	$Always$	$Poor$	$Average$	No	$Fail$
$TengTong$	$Sometimes$	$Poor$	$Average$	Yes	$Pass$
$QuocPhong$	$Rarely$	$Average$	$Average$	Yes	$Pass$
$Sreejith$	$Rarely$	$Good$	$Good$	Yes	$Pass$
$Mason$	$Rarely$	$Good$	$Good$	No	$Fail$
$Lucas$	$Sometimes$	$Good$	$Good$	No	$Pass$
$Rishav$	$Always$	$Average$	$Average$	Yes	$Fail$
$Ryutaro$	$Always$	$Good$	$Good$	Yes	$Pass$
$Zhongxiang$	$Rarely$	$Average$	$Good$	Yes	$Pass$
$Yehong$	$Always$	$Average$	$Good$	No	$Pass$
$Yizhou$	$Sometimes$	$Average$	$Average$	No	$Pass$
$Dmitrii$	$Sometimes$	$Poor$	$Good$	Yes	$Pass$
$Rachael$	$Rarely$	$Average$	$Average$	No	$Fail$

Table 3: Positive ($MLGrade = Pass$) and negative ($MLGrade = Fail$) training examples for target concept $MLGrade$.

1. (5 points) What is the entropy of the target concept '$MLGrade$' on all the 14 training examples shown in Table 3 above? Give your answer up to 3 decimal places.

Solution:
$H(MLGrade) =$

What is the information gain with choosing the input attribute '$AttendClass$' as the root of the decision tree? Give your answer up to 3 decimal places.

Solution:
$Gain(MLGrade, AttendClass) =$

What is the information gain with choosing the input attribute '$MidtermGrade$' as the root of the decision tree? Give your answer up to 3 decimal places.

Solution:
$Gain(MLGrade, MidtermGrade) =$

2. (5 points) Using on the DECISION-TREE-LEARNING algorithm given on page 10 of "Decision Tree Learning" lecture slides (reproduced in Fig. 2) and information gain as the IMPORTANCE function in this algorithm, draw the resulting decision tree that is induced by all the 14 training examples shown in Table 3 above. You are required to label the non-leaf nodes with the input attributes, the leaf nodes with the ML grade (i.e. either '$Pass$' or '$Fail$'), and the branches with the values of the chosen attributes.

```
function DECISION-TREE-LEARNING(examples, attributes, parent_examples) returns
tree

  if examples is empty then return PLURALITY-VALUE(parent_examples)
  else if all examples have the same classification then return the classification
  else if attributes is empty then return PLURALITY-VALUE(examples)
  else
     A ← argmax_{a ∈ attributes} IMPORTANCE(a, examples)
     tree ← a new decision tree with root test A
     for each value v_k of A do
        exs ← {e : e ∈ examples and e.A = v_k}
        subtree ← DECISION-TREE-LEARNING(exs, attributes − A, examples)
        add a branch to tree with label (A = v_k) and subtree subtree
     return tree
```

Figure 2: DECISION-TREE-LEARNING algorithm.

Solution:

3. (2 points) Give the logical expression (in disjunctive normal form) that corresponds to the decision tree produced by the DECISION-TREE-LEARNING algorithm. Let '$Pass$' and '$Fail$' be associated with True and False, respectively. An example of a logical expression (in disjunctive normal form) is

$$(AttendClass = Sometimes) \ \vee \ (ProjectGrade = Average \wedge LoveML = Yes) \ .$$

Solution:

Part IV
Neural Networks

(13 points) Structured questions. Answer in the space provided on the script.

1. (13 points) Cara has constructed a dataset of 4 Boolean input attributes x_1, x_2, x_3, x_4 and 4 Boolean target outputs $t_{k_1}, t_{k_2}, t_{k_3}, t_{k_4}$ with the following 4 training examples of the form $d = \langle (x_1, x_2, x_3, x_4), (t_{k_1}, t_{k_2}, t_{k_3}, t_{k_4}) \rangle$:

$$D = \{d_1 = \langle (1,0,0,0), (1,0,0,0) \rangle, d_2 = \langle (0,1,0,0), (0,1,0,0) \rangle, \\ d_3 = \langle (0,0,1,0), (0,0,1,0) \rangle, d_4 = \langle (0,0,0,1), (0,0,0,1) \rangle\}.$$

Construct and draw a network of perceptron units with **only a hidden layer of two units** based on the following **constraints**:

- There should be **only four (Boolean) output units** k_1, k_2, k_3, k_4 and **four input units** (i.e., one input unit for each of the four (Boolean) input attributes x_1, x_2, x_3, x_4).
- A Boolean is **0** if false, and **1** if true.
- The activation function $\sigma(\cdot)$ of every (non-input) unit is a **0** to **1** step function, including that of the output unit. That is,

$$\sigma(s) = \begin{cases} 1 & \text{if } s > 0, \\ 0 & \text{otherwise.} \end{cases}$$

  This is in contrast to the activation function on page 6 of the "Neural Networks" lecture slides that is a -1 to 1 step function.
- **Your weights (i.e., hypothesis) must be consistent with D and must take on one of the following values: –1, 0, 1.**
- A bias input is of value 1 and is not considered a hidden unit.
- You don't have to draw edges with weight 0.

Solution:

END OF PAPER